**PROBLEM SET 5, PART 2: LIE GROUPS ACTIONS**
**DUE: NOV. 28**

(1) [**The Lie algebra $\mathfrak{so}(3)$**]
(a) Prove: The Lie algebra $\mathfrak{so}(3)$ has a basis

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Moreover, prove it is isomorphic to the Lie algebra $(\mathbb{R}^3, \times)$, where $\times$ is the cross product.
(b) Consider the $\mathrm{SO}(3)$-action on $\mathbb{R}^3$ by rotations. Prove: The induced vector fields associated to $A, B, C$ on $\mathbb{R}^3$ are

$$x^3\frac{\partial}{\partial x^2} - x^2\frac{\partial}{\partial x^3}, \qquad x^1\frac{\partial}{\partial x^3} - x^3\frac{\partial}{\partial x^1}, \qquad x^2\frac{\partial}{\partial x^1} - x^1\frac{\partial}{\partial x^2}.$$

(2) [**Invariants of a Lie group action**] [Not required]
Suppose $G$ acts on $M$ smoothly.
(a) Prove: The formula

$$(g \cdot \varphi)(x) := \varphi(g^{-1} \cdot x)$$

defines a linear action of $G$ on $C^\infty(M)$.

**Definition 0.1.** A fixed point of the above induced action of $G$ on $C^\infty(M)$ is called an *invariant* of the $G$-action on $M$. The set of invariants is denoted by $C^\infty(M)^G$.

(b) Prove: If $G$ is connected, then $f \in C^\infty(M)^G$ if and only if $X_M f = 0$ for all $X \in \mathfrak{g}$.
(c) Suppose $G$ acts on $M$ properly and freely. Prove: $C^\infty(M)^G \simeq C^\infty(M/G)$.
(d) What are the invariants for the standard $\mathrm{O}(n)$ action on $\mathbb{R}^n$?
(e) What are the invariants for the adjoint action of $\mathrm{GL}(n, \mathbb{C})$ on $\mathfrak{gl}(n, \mathbb{C})$?

(3) [**More Lie subgroups/subalgebras** ]
Let $G$ is a Lie group, with Lie algebra $\mathfrak{g}$.
(a) Define the center of $G$ to be $Z(G) = \{g \in G \mid gg' = g'g \text{ for all } g' \in G\}$. Show that $Z(G)$ is a Lie subgroup of $G$, and find its Lie algebra.
(b) Let $\varphi : G \to G$ be a Lie group homomorphism. Let $H$ be the set of $\varphi$-fixed points, i.e. $H = \{g \in G \mid \varphi(g) = g\}$. Prove that $H$ is a Lie subgroup of $G$, and find its Lie algebra.

(4) [**Equivariant maps**] [Not required]
Let $M, N$ be $G$-manifolds. A smooth map $f : M \to N$ is called $G$-equivariant if

$$g \cdot f(m) = f(g \cdot m), \forall g \in G, m \in M.$$

Prove:

(a) Prove: If $G$ acts transitively on $M$, then $f$ is of constant rank.
(b) Suppose $G$ is connected, then $f$ is $G$-equivariant if and only if it is $\mathfrak{g}$-equivariant in the sense

$$X_N(f(m)) = df_m(X_m), \forall X \in \mathfrak{g}, m \in M.$$